Isaac Alexander Vazquez

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Education

California Polytechnic State University — *San Luis Obispo, CA*
Bachelor of Science in Computer Engineering — *Expected May 2027*

Skills

- **Programming languages:** C, Python, SystemVerilog, RISC-V Assembly
- **Hardware Platforms:** FreeRTOS, Linux, ROS2 Humble, Oscilloscope, Multimeter, GazeboSim, Foxglove
- **Microcontrollers:** STM32 ARM-Cortex Microcontrollers, NVIDIA Jetson Nano Dashboard, ESP32, Basys3Board
- **Languages:** Spanish (fluent)

Projects

Amiga Robot Driver Configurations | *ROS2, NVIDIA Jetson, CAN Bus* — *Aug 2026*

- Joined an existing four-wheeled Agrivulture robotics project and re-evaluated drivers built by a previous team to assess performance.
- Sniffed CAN packets between a PC and a Jetson Nano, analyzing frames with ROS2 Linux commands and oscilloscope measurements of voltage changes.
- Documented debugging processes for future collaborators and created launch files to start multiple ROS2 nodes concurrently.

RTOS Traffic Light System | *FreeRTOS, Embedded Systems, STM32* — *June 2026*

- Re-architected a bare-metal traffic light system into a multi-task FreeRTOS application handling button input and asynchronous interrupts.
- Replaced polling and blocking delays with interrupt-driven button handling, using FreeRTOS APIs for precise timing and inter-task resource management.
- Debugged over UART using PuTTY and used software timers with callbacks to reduce task overhead.

Function Generator | *Embedded Systems, Timer Interrupts, SPI, STM32* — *Oct 2025*

- Designed and debugged an STM32L476RG system that generates AC waveforms with user-selectable shape, duty cycle, and frequency.
- Implemented SPI communication (master mode) to drive an external 12-bit DAC for waveform output.
- Achieved 100-500 Hz output from keypad input using polling and timer interrupts, verified on an oscilloscope.

RISC-V OTTER Microcontroller with Interrupts | *FPGA, Vivado, RISC-V Assembly* — *May 2024*

- Designed and debugged a 32-bit RISC-V (RV32I) processor in SystemVerilog using Vivado, including an ALU module, Program Counter and Control Decoder Finite State machine running RISC-V Assembly programs
- Implemented interrupt support using CSR registers and an MRET instruction, replacing polling so the CPU can service button and switch events without wasting clock cycles
- Verified with waveform-simulation in Vivado, then tested on a Xilinx Basys3 board with custom Assembly programs, such as a switch-activated LED counter

Employment

ITS Student Assistant — *San Luis Obispo, CA*
Cal Poly Information Technology Services — *May 2025 – Present*

- Provide technical support to students, faculty, and staff across hardware, software, and network issues, resolving about 10 issues per week
- Image and deploy 5+ Windows and macOS devices for faculty and staff, installing applications and configuring campus network connectivity IPV4.
- Document troubleshooting steps from client tickets in Jira, creating a reference used by student assistants and for training new hires